ESCO-LUCE: Confidence-Aware Semantic Recommendation of Adaptive Learning Paths

Antonio Ferrara[1], Giuseppe Murgolo[1], Gabriele Squeo[1] and Tommaso Di Noia[1]

[1]*Politecnico di Bari, Bari, Italy*

Abstract

Professional upskilling requires retrieving and sequencing learning materials that resolve authentic skill deficits without treating unverified abilities as absent. We present ESCO-LUCE, a semantic recommendation framework that operationalizes European ESCO concepts for adaptive curriculum planning. By separating estimated mastery from evidentiary confidence, the framework distinguishes actual instructional needs from competencies that appear sufficient but remain weakly supported. The latter are verified through targeted, resource-grounded assessments before instructional decisions are finalized. Verified mastery deficits are passed to an A^* planner that retrieves prerequisite-compliant, duration-minimizing module sequences, while new assessment evidence triggers replanning of the unfinished path suffix. Two controlled offline evaluations show that, under binary skill coverage, the planner consistently recovers the minimum-duration solutions returned by uniform-cost search while exploring substantially fewer states. Under noisy learner estimates, the assessment-first policy nearly matches the deficit recall of a conservative policy that converts uncertainty directly into instruction, while recommending substantially less instructional time. These findings support confidence-aware verification as an effective alternative to assigning coursework solely because prior competence evidence is incomplete.

Keywords

adaptive learning, educational information retrieval, knowledge graphs, learning paths, learner modeling, ESCO

1. Introduction

The rapid evolution of job roles and required skills has made workforce upskilling an increasingly important challenge [1]. Professionals can now choose from a large and growing number of learning resources, but finding suitable material is only part of the problem. They also need to understand which resources match their current skills, which ones address the competences required for a target role, and in what order they should be studied. Workforce upskilling can thus be viewed as a goal-directed sequential retrieval problem.

Standard recommendation algorithms are not well suited to this setting. Collaborative filtering and content-based methods typically rank learning resources according to user preferences, predicted relevance, or similarity with previously selected content [2, 3, 4]. In professional training, however, learning resources are often connected by prerequisite relationships. A module may be highly relevant to a target job profile but still be unsuitable if the learner lacks the required background knowledge. Recommending introductory material on concepts that the learner has already mastered is equally inefficient. The problem is not only to identify relevant modules, but also to organize them into a sequence that respects prerequisites and covers the missing competencies with as little unnecessary training as possible.

Knowledge graphs provide a useful way to connect course descriptions, job requirements, and the skills they refer to [5, 6, 7]. In Europe, the ESCO classification offers a standardized and multilingual vocabulary for occupations and skills [8], allowing courses and job profiles to be represented using a common set of concepts. ESCO alone, however, does not address several practical aspects of curriculum generation. It does not define how uncertainty about a learner's current skills should be represented, how assessment results should update this information, or how courses should be selected under time and prerequisite constraints [9, 10].

A related issue concerns the way missing skills are identified. In the absence of explicit historical records, competence-based recommenders often rely on a closed-world simplification, for example by

Proceedings of the 16th Italian Information Retrieval Workshop, IIR 2026, November 12–14, 2026, Rome, Italy



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defining skill gaps through set difference [9, 10]. This effectively treats unverified competencies as missing competencies and may lead to redundant coursework for learners who already possess the required knowledge.

In this paper, we introduce **ESCO-LUCE**, an adaptive recommendation framework based on ESCO concepts. The framework uses ESCO identifiers as shared references for skills and occupations, while user-specific and application-specific information is represented through separate context nodes. Relationship reification is used to model target profiles, module prerequisites, learning materials, and learner states without modifying the original ESCO taxonomy. The learner model distinguishes between *mastery*, which represents the estimated level of knowledge for a skill, and *confidence*, which represents how strongly that estimate is supported by available evidence. Keeping these two quantities separate allows the system to address two complementary problems:

- **Targeted diagnostic retrieval:** the system identifies skills for which estimated mastery is high but confidence is low, and retrieves related textual resources to support the generation of diagnostic questions;
- **Time-sensitive path planning:** an A^* planner selects a sequence of modules that covers confirmed skill gaps while minimizing study time and respecting prerequisite constraints.

When a new assessment result becomes available, the learner model is updated through an evidence-weighted fusion rule. The system then replans only the remaining part of the learning path. Completed modules are left unchanged, while future modules can be reordered or removed according to the updated learner state.

The main contributions of this work are:

1. an operational ontology that links curriculum targets, module prerequisites, learning resources, and assessment records to ESCO skills without modifying the underlying taxonomy;
2. a learner model that represents mastery and confidence separately, together with an evidence-fusion rule that updates both quantities from assessment results;
3. a time-sensitive A^* planning method that recomputes the remaining learning path after each assessment update;
4. an empirical evaluation on 270 synthetic catalogs and 1,000 simulated learner profiles, measuring search scalability and the effect of confidence-based assessment on redundant study time.

2. Related Work

2.1. Semantic Models in Adaptive Learning

Adaptive educational systems need to connect workplace skill requirements with available learning resources. Latent embeddings and vector similarity are useful for matching text, but they do not directly represent prerequisites, proficiency thresholds, or the reasoning behind a recommended learning path. Semantic Web standards such as RDF, OWL, and SKOS [11, 12, 13, 14] provide an explicit relational structure that can be used to represent these constraints and make recommendation decisions easier to inspect.

The separation between domain knowledge, learner information, and pedagogical rules has a long tradition in adaptive hypermedia [15]. More recent ontology-based recommenders follow a similar approach, for example by representing courses as collections of knowledge units and incorporating diagnostic feedback into the recommendation process [16]. At the same time, reviews of ontology-based learning systems highlight several practical limitations. Complex schemas can make graph traversal expensive, learner-state updates are often based on system-specific heuristics, and the adopted vocabularies do not always align well with terminology used in real occupational settings [9, 10].

This last issue is especially relevant in workforce training, where the same skill may be described differently across job postings, universities, and training providers. ESCO addresses part of this problem by providing shared URIs for occupations and skills across European languages [8]. Initiatives such as TalentCLEF also show that mapping unstructured text to ESCO skills is itself an important Information Retrieval task [17]. In this work, we assume that this mapping has already been performed and focus

on the next step: using the resulting skill identifiers to construct learning paths that account for prerequisites, time constraints, and uncertainty about the learner's current knowledge.

2.2. Learning Path Construction as Sequential Retrieval

Standard recommender systems usually rank individual items according to predicted relevance or user preference [3, 4], and are commonly evaluated with metrics such as Precision, Recall, and nDCG [18]. This formulation does not fully capture the requirements of professional training. Content-based methods do not naturally account for prerequisite relationships and may recommend multiple resources covering the same skills. Collaborative filtering can be difficult to apply in specialized catalogs, where interaction data are often sparse and new users or courses have little history. Session-based approaches, in turn, are generally designed to predict short-term interaction patterns rather than to construct a complete learning path toward a predefined competence target.

Curriculum construction is better viewed as a sequential retrieval problem in which the usefulness of each module depends on the learner's current state and on the modules selected before it. Three aspects are particularly important. First, some modules can only be taken once specific prerequisite skills have been acquired. Second, repeatedly covering a skill that is already mastered provides little or no additional benefit. Third, the objective is not simply to rank relevant resources, but to reach the target set of competencies while keeping the overall training time as low as possible.

Knowledge graphs have already been used to study career transitions and reskilling pathways [19, 6], but selecting a complete curriculum remains a combinatorial search problem. We formulate path construction as an A^* search over competence-gap states [20]. Each state represents the skills that are still missing from the target profile, while the heuristic estimates a lower bound on the remaining study cost. In the binary-coverage setting considered in this work, this heuristic is admissible and therefore preserves the optimality guarantees of A^*.

3. Semantic Modeling and Grounding Pipeline

Using an external taxonomy such as ESCO in an adaptive learning system requires a clear separation between shared semantic concepts and application-specific information. In ESCO, a skill is represented as a stable concept. Within a learning system, however, the same skill can play different roles: it may define a requirement for a target occupation, act as a prerequisite for a module, represent a learning outcome, or describe the current competence of a learner. Storing learner-specific or course-specific information directly on ESCO nodes would mix these different contexts and make the shared taxonomy harder to reuse consistently.

ESCO-LUCE therefore keeps ESCO entities unchanged and uses them only as external references. Application-specific information is represented through a separate operational schema, defined in OWL/RDF and materialized as a Neo4j property graph. Figure 1 presents a simplified view of the ontology, limited to the classes and relations most relevant to the workflow described in this paper[1].

3.1. Operational Ontology and Reified Learner State

As shown in Figure 1, relations between catalog entities and skills are represented through classes derived from an abstract `SkillRelation`. Each relation refers to a canonical `esco:Skill` entity through the `refersToSkill` predicate. The schema is built around four main entities:

- **Occupational Profile (`Profile`):** represents the target job role. A profile is connected through `SkillProfile` relations to the target skills $s \in S_p$, each associated with a required proficiency threshold $t_s \in [0, 1]$.
- **Learning Module (`Module`):** represents a unit of instruction with an estimated duration $d(M)$, expressed in minutes. Modules are connected to skills through `TaughtSkill` relations, which

[1]The complete ontology is publicly available at https://split.to/esco-luce-owl.

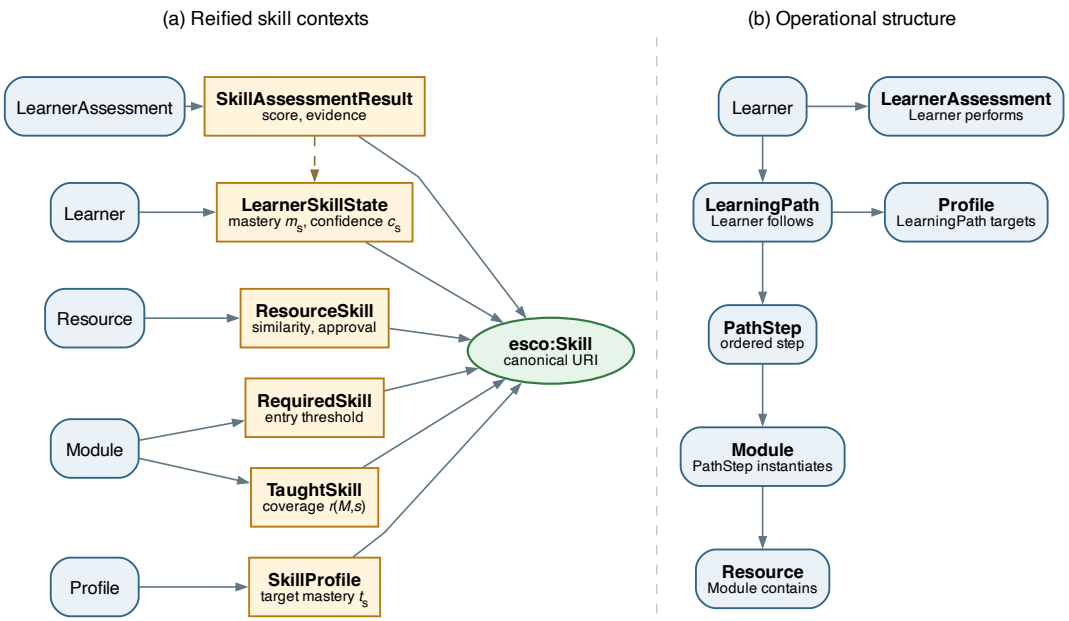


Figure 1: Reified skill contexts in the ESCO-LUCE operational knowledge model. Blue rounded nodes denote application entities, orange rectangular nodes denote relation classes, and the green ellipse represents the canonical external ESCO concept. Each reified relation refers to `esco:Skill` through `refersToSkill`.

specify a coverage weight $r(M, s) \in [0, 1]$, and through `RequiredSkill` relations, which represent hard prerequisites.

- **Educational Resource (`Resource`):** represents a textual learning asset contained in a module. Resources are linked to the skills they address through `ResourceSkill` relations.
- **Curriculum Trajectory (`LearningPath` and `PathStep`):** represents the ordered learning path assigned to a learner. Each step refers to a module and records its current execution status.

Separating Current Learner State from Assessment History The model keeps the learner's current skill estimates separate from the assessment records that produced them:

1. `LearnerSkillState` stores the current state associated with skill s, including estimated mastery $m_s \in [0, 1]$, confidence $c_s \in [0, 1]$, and the corresponding timestamps. The planner reads these values directly when constructing or updating a learning path.
2. `SkillAssessmentResult` stores the outcome of an individual assessment event associated with a `LearnerAssessment`. It records the questions presented to the learner, their cognitive difficulty, the submitted responses, and the resulting scores.

This separation preserves the assessment history while keeping the current learner state directly accessible to the planning algorithms. As a result, each pair $\langle m_s, c_s \rangle$ can be traced back to the assessment records used to derive it.

3.2. Semantic Ingestion and Skill Linking

Unstructured educational materials (e.g., lecture notes, syllabi, and technical documentation) enter the catalog through an automated linking stage that connects text segments to canonical ESCO skills.

For an extracted text passage x_j and skill metadata z_s, a dense sentence encoder $e(\cdot)$ maps both inputs into a common semantic embedding space [21]:

$$\hat{s}_j = \arg \max_{s \in \mathcal{S}_{\text{ESCO}}} e(x_j)^\top e(z_s). \tag{1}$$

Our implementation uses normalized embeddings from `paraphrase-multilingual-mpnet-base-v2`, retains the highest-scoring ESCO candidate for each segment when cosine similarity exceeds 0.55, and

presents the resulting associations for human confirmation. Approved associations are persisted as reified relation nodes in a Neo4j property graph, allowing the downstream planner to query modules with verified durations $d(M)$, prerequisites, and skill coverage weights $r(M, s)$.

3.3. Content Grounding and Diagnostic Question Synthesis

In order to avoid relying on static question banks that quickly become misaligned with evolving catalog materials, diagnostic evaluation items are synthesized dynamically from the educational resources linked to the target skill:

1. **Targeted Resource Retrieval:** When the diagnostic policy flags a skill as requiring verification, the service retrieves from ChromaDB the textual resource explicitly linked to that skill and module.
2. **Conditioned Item Generation:** The retrieved resource, canonical ESCO descriptions, and a target cognitive level from Bloom's revised taxonomy form a prompt for a generative language model, producing a multiple-choice item with one resource-grounded correct answer and three plausible distractors.
3. **Deterministic Scoring:** Learner responses are scored against the reference key, producing an immutable `SkillAssessmentResult` record that triggers the mathematical state update rules.

Restricting the language model to question formulation while delegating state tracking, prerequisite checking, and path search to the deterministic graph engine limits the propagation of generative errors into the planning logic.

4. Confidence-Aware Recommendation and Replanning

The recommendation process is iterative. Starting from the learner's current state, the system identifies missing competencies, plans a sequence of modules through A^* search, and uses targeted assessments when additional evidence about a skill is needed.

4.1. Bivariate State and Dual Gap Formulation

For each skill s required by target profile p ($s \in S_p$), the system retrieves the target mastery level t_s from `SkillProfile` and the learner's current mastery m_s and confidence c_s from `LearnerSkillState`. Given a target confidence threshold $c^* \in (0, 1]$, we define mastery gap $g_m(s)$ and confidence gap $g_c(s)$:

$$g_m(s) = \max(0,\, t_s - m_s), \tag{2}$$

$$g_c(s) = \max(0,\, c^* - c_s). \tag{3}$$

Mastery and confidence are then used in two different parts of the recommendation process:

1. **Confidence-weighted path signal (g_{path}):** For skills with a remaining mastery gap, the planner combines the size of the gap with the confidence associated with the current estimate:

$$g_{\text{path}}(s) = \mathbb{I}[g_m(s) > \epsilon]\,(\alpha g_m(s) + (1-\alpha) g_c(s))\,, \tag{4}$$

   where $\mathbb{I}[\cdot]$ is the indicator function and $\alpha \in [0, 1]$ controls the relative contribution of mastery and confidence. We use $\alpha = 0.7$ as the main operating point in the experiments. The tolerance $\epsilon = 0.05$ is used to ignore small residual mastery gaps and consider the corresponding skill sufficiently covered.
2. **Diagnostic assessment priority (q):** Assessment is instead used for skills whose estimated mastery already meets the target but is supported by insufficient evidence:

$$q(s) = \mathbb{I}[m_s \geq t_s]\, g_c(s). \tag{5}$$

The two quantities therefore have different purposes. The planner considers skills that still have a mastery gap, with g_{path} assigning greater weight to cases in which the current estimate is less certain. The diagnostic score $q(s)$ is used when the learner appears to satisfy the target requirement but confidence is still too low to justify skipping instruction without further verification.

4.2. Time-Sensitive A^* Path Synthesis

The search state is defined by the residual mastery-gap vector $\mathbf{g}_{m,n} = [g_{m,n}(s)]_{s\in S_p}$. Confidence remains fixed during each planning episode and is updated only when new assessment evidence becomes available. At node n, the planner computes:

$$g_{\text{path},n}(s) = \mathbb{I}[g_{m,n}(s) > \epsilon]\,(\alpha g_{m,n}(s) + (1-\alpha)g_c(s))\,, \tag{6}$$

which weights open mastery gaps according to the confidence of the current estimate. The contribution of a candidate module M is:

$$P(M \mid n) = \sum_{s\in S_p} r(M,s)\, g_{\text{path},n}(s), \tag{7}$$

with efficiency $E(M \mid n) = P(M \mid n)/d(M)$.

Path construction is formulated as an A^* search. Each node contains the current module sequence, its cumulative duration $G(n) = \sum_{M\in\text{path}(n)} d(M)$, and $\mathbf{g}_{m,n}$. A module is eligible only if its prerequisites are satisfied by the learner or by earlier modules in the path. Adding module M updates the mastery gaps as

$$g_{m,n+1}(s) = \max\left(0,\, g_{m,n}(s) - r(M,s)\right), \tag{8}$$

while confidence remains unchanged and $g_{\text{path},n+1}$ is recomputed. This transition is a predictive rollout internal to the search and does not modify the persistent `LearnerSkillState`. Stored mastery and confidence are updated only after new assessment evidence is observed.

The heuristic estimates the remaining duration as:

$$H(n) = \frac{\sum_{s\in S_p} g_{\text{path},n}(s)}{E_{\max}(n)}, \qquad F(n) = G(n) + H(n), \tag{9}$$

where $E_{\max}(n)$ is the highest efficiency among modules not yet selected. Prerequisites are ignored only when computing $H(n)$ and remain enforced during node expansion. In the binary-coverage setting of RQ1, with $r(M,s) \in \{0,1\}$, this relaxation does not overestimate the remaining study time, so the heuristic is admissible. The search terminates when $g_{m,n}(s) \le \epsilon$ for all $s \in S_p$, yielding a minimum-duration path. For fractional or stochastic gains, the same heuristic guides the search, but admissibility is not assumed.

4.3. Evidence-Weighted State Fusion and Suffix Replanning

When a diagnostic test or post-module quiz is administered, each item evaluates one or more target skills. Question i is assigned a cognitive difficulty level $d_i \in \{1,\ldots,6\}$ according to Bloom's taxonomy [22]. Given the correctness indicator $y_i \in \{0,1\}$, its weight is defined as:

$$w_i = 1 + \lambda d_i, \tag{10}$$

so that questions requiring higher cognitive effort contribute more to the assessment. For the items associated with skill s, test mastery and test confidence are computed as:

$$m_{\text{test}} = \frac{\sum_i w_i y_i}{\sum_i w_i}, \tag{11}$$

$$c_{\text{test}} = 1 - \exp\left(-\frac{\sum_i w_i}{\tau}\right), \tag{12}$$

where $\tau > 0$ controls how quickly confidence grows with the amount of assessment evidence [23]. The exponential form introduces diminishing returns: the first questions have a stronger effect on confidence, while additional evidence produces progressively smaller increases.

The current learner state is then updated as

$$m' = c_{\text{test}} m_{\text{test}} + (1 - c_{\text{test}}) m, \tag{13}$$

$$c' = \min\left(1,\, c + c_{\text{test}}(1 - c)\right). \tag{14}$$

The mastery update moves the current estimate toward the observed test performance according to the confidence of the new evidence. The confidence update accumulates new evidence while keeping c' bounded in $[0, 1]$.

After the update, only the unfinished portion of the learning path is replanned. Completed modules remain unchanged, the residual mastery gaps are recomputed from the updated learner state, and A^* is run again on the remaining curriculum. Future modules may therefore be removed when the updated state shows that their target skills are already covered, or added when relevant gaps remain.

4.4. Illustrative Example

This example illustrates how learner-state updates affect the planned curriculum. Consider a learner targeting the profile `Junior Data Analyst`, which requires SQL Querying (s_1) and Python Data Processing (s_2), both with target mastery $t_s = 0.80$ and confidence target $c^* = 0.80$. The initial learner state is $(m_{s_1}, c_{s_1}) = (0.45, 0.60)$ for SQL and $(m_{s_2}, c_{s_2}) = (0, 0)$ for Python.

The catalog contains three modules: M_1 (Relational Databases with SQL, 120 min), which covers s_1; M_2 (Python for Data Analysis, 180 min), which covers s_2 and requires SQL mastery of at least 0.80; and M_{rem} (SQL Remedial Lab, 45 min), which provides additional SQL coverage after M_1 when a residual gap remains.

Initial Planning With $\alpha = 0.7$, the initial SQL gaps are $g_m(s_1) = 0.35$ and $g_c(s_1) = 0.20$, giving $g_{\text{path}}(s_1) = 0.305$. Since $m_{s_1} < t_{s_1}$, $q(s_1) = 0$: SQL requires instruction rather than pre-instruction verification. For Python, $g_m(s_2) = g_c(s_2) = g_{\text{path}}(s_2) = 0.80$.

Module M_2 cannot yet be selected because its SQL prerequisite is not satisfied. Under the expected mastery transition, M_1 closes the SQL gap and unlocks M_2, producing the initial path $\langle M_1, M_2 \rangle$ with a total duration of 300 minutes.

Assessment and Suffix Replanning After M_1, the system generates five SQL questions at Bloom levels $(2, 3, 3, 4, 4)$. With $\lambda = 0.2$, four correct answers yield $m_{\text{test}} \approx 0.780$ and, for $\tau = 5.0$, $c_{\text{test}} \approx 0.806$. Applying the fusion rules gives

$$m'_{s_1} \approx 0.716, \qquad c'_{s_1} \approx 0.922. \tag{15}$$

The assessment therefore provides strong evidence that SQL mastery is still slightly below the required level. The confidence gap is now zero, while $g_m(s_1) = 0.084$, giving an updated priority $g_{\text{path}}(s_1) \approx 0.059$.

Since the SQL prerequisite of M_2 is still not satisfied, the completed prefix $\langle M_1 \rangle$ is preserved and only the remaining path is replanned. The original suffix $\langle M_2 \rangle$ becomes $\langle M_{\text{rem}}, M_2 \rangle$. Under the expected mastery transition, M_{rem} closes the residual SQL mastery gap and makes M_2 eligible; confidence remains unchanged until new assessment evidence is collected.

This example shows how assessment evidence can introduce targeted remediation without recomputing or altering the completed part of the learning path.

5. Experimental Evaluation and Results

We evaluate ESCO-LUCE through two controlled offline experiments addressing the following research questions:

Table 1
Curriculum planning performance across 270 benchmark catalogs (mean ± 95% CI).

Search Method	Duration (min)	Regret (%)	Redundancy (min)	Node Expansions
UCS Oracle	343.8 ± 11.1	0.0	24.1 ± 2.9	3941.5 ± 998.1
ESCO-LUCE A^*	$\mathbf{343.8 \pm 11.1}$	**0.0**	$\mathbf{23.8 \pm 2.9}$	$\mathbf{461.7 \pm 112.3}$
Greedy Efficiency	365.1 ± 12.4	6.0 ± 0.8	27.8 ± 3.1	127.7 ± 10.8
Shortest-First	531.9 ± 19.3	53.8 ± 1.8	38.0 ± 4.2	289.8 ± 24.7

- **RQ1:** Can the prerequisite-aware A^* planner recover minimum-duration curricula, and how much search effort does it save compared with uninformed search and greedy heuristics?
- **RQ2:** Under noisy competence estimates, can confidence-aware diagnostic assessment reduce false skips without introducing excessive instructional overhead?

The two experiments separately examine path-search performance and decision making under uncertain learner states. Reported results are means with 95% confidence intervals computed as $1.96s/\sqrt{n}$.

5.1. Combinatorial Curriculum Benchmark (RQ1)

Experimental Setup We generate 270 curriculum catalogs from a balanced grid of target profile sizes $|S_p| \in \{8, 12, 16\}$ and prerequisite probabilities $p_{\text{pre}} \in \{0, 0.15, 0.30\}$, with 30 independent instances for each configuration. Catalog size increases with target complexity, using $|M| \in \{20, 40, 70\}$ modules. Prerequisites form directed acyclic graphs, and each target skill is covered by at least one single-skill module to ensure that a feasible solution exists. Multi-skill modules cover between two and six skills and are assigned sub-additive durations to model the time advantage of integrated instruction. Each target skill is independently marked as already mastered with probability 0.20.

We compare ESCO-LUCE against three baselines:

- **Uniform-Cost Search (UCS):** an uninformed optimal search that expands states in increasing order of cumulative duration and provides the reference minimum cost;
- **Greedy Efficiency:** selects at each step the eligible module with the highest coverage-to-duration ratio $E(M)$;
- **Shortest-First:** selects the shortest eligible module that covers at least one remaining skill gap.

Results and Discussion Table 1 reports the results across the 270 benchmark instances. All methods produced paths that respected the generated prerequisite constraints.

ESCO-LUCE A^* reached the same minimum duration as UCS in every instance, resulting in zero regret. The difference lies in search effort: UCS required an average of 3941.5 node expansions, compared with 461.7 for A^*. This corresponds to an average per-instance reduction of 83.9% in explored states.

The greedy methods required fewer expansions but produced longer curricula. Greedy Efficiency incurred an average regret of 6.0%, corresponding to 21.3 additional minutes. Its local coverage-to-duration criterion can favor modules that appear efficient early in the path but later overlap with modules needed to satisfy prerequisites. Shortest-First performed substantially worse, with 53.8% regret and an average of 38.0 redundant minutes. Selecting short modules independently tends to fragment the curriculum and misses the time savings offered by modules that cover several required skills together.

5.2. Noisy Learner Simulation Protocol (RQ2)

Experimental Setup We next simulate 1,000 learners targeting a profile with $|S_p| = 12$ skills. Target mastery is fixed at $t_s = 0.70$ and target confidence at $c^* = 0.80$. True learner mastery is sampled as $m_s^* \sim \text{Beta}(2.2, 2.0)$, while initial confidence follows $c_s \sim \text{Beta}(1.8, 1.8)$.

Table 2
Policy performance across 1,000 simulated learners under noisy estimation (mean ± 95% CI).

Policy	Instruction (min)	Questions	False Skips	Target Recall
Mastery-Only	$\mathbf{582.9 \pm 7.2}$	$\mathbf{0.0}$	1.39 ± 0.07	$.848 \pm .008$
Joint Gap ($\alpha = 0.7$)	729.2 ± 5.2	0.0	$\mathbf{0.43 \pm 0.04}$	$\mathbf{.953 \pm .004}$
Assess-Then-Plan	672.8 ± 6.3	15.3 ± 0.5	0.47 ± 0.04	$.948 \pm .005$

To represent uncertainty in background information such as self-reported skills or previous coursework, the system observes a noisy mastery estimate:

$$\hat{m}_s = \text{clip}\left(m_s^* + \epsilon_s,\, 0,\, 1\right), \qquad \epsilon_s \sim \mathcal{N}(0, \sigma_s^2), \tag{16}$$

with:

$$\sigma_s = 0.06 + 0.30(1 - c_s). \tag{17}$$

Low-confidence estimates are therefore more strongly perturbed, with σ_s ranging from approximately 0.36 at very low confidence to 0.06 when confidence is high.

RQ2 focuses on the recommendation policy rather than path construction. Each skill is therefore associated with one instructional unit whose duration is sampled uniformly between 45 and 90 minutes; multi-module sequencing is evaluated separately in RQ1. Diagnostic tests contain five items at Bloom levels $(2, 3, 3, 4, 4)$. For an item with difficulty d_i, correctness is sampled with probability:

$$\text{clip}(m_s^* + 0.18 - 0.06d_i,\, 0,\, 1). \tag{18}$$

We use $\lambda = 0.2$ for item weighting and $\tau = 5.0$ for confidence estimation.

Three recommendation policies are compared:

- **Mastery-Only:** assigns instruction whenever the observed mastery estimate is below target, $\hat{m}_s < t_s$, without considering confidence;
- **Joint Gap ($\alpha = 0.7$):** combines mastery and confidence gaps as

  $$g_{\text{joint}}(s) = \alpha g_m(s) + (1 - \alpha) g_c(s),$$

  and assigns instruction whenever $g_{\text{joint}}(s) > 0.05$. Confidence alone can therefore trigger instruction;
- **Assess-Then-Plan (ESCO-LUCE):** tests skills for which estimated mastery meets the target but confidence remains below c^*, $\hat{m}_s \geq t_s \wedge c_s < c^*$, and uses the resulting evidence before deciding whether instruction is needed.

We measure instructional duration, number of diagnostic questions, *false skips*—true deficits with $m_s^* < t_s$ that are incorrectly omitted from instruction—and *target recall*, defined as the proportion of true deficits selected for instruction.

Results and Discussion As shown in Table 2, Mastery-Only recommends the least instruction, with an average of 582.9 minutes, but is more sensitive to noisy estimates. It produces 1.39 false skips per learner and a target recall of 0.848 because apparently sufficient mastery is accepted even when the supporting evidence is weak.

Joint Gap substantially reduces this problem, reaching 0.953 recall and 0.43 false skips per learner. The improvement comes at the cost of additional instruction: the policy recommends 729.2 minutes on average, 146.3 minutes more than Mastery-Only. This occurs because low confidence can itself trigger coursework even when the estimated mastery already satisfies the target.

Assess-Then-Plan reaches a similar level of coverage without treating uncertainty directly as an instructional deficit. With an average of 15.3 diagnostic questions per learner, it achieves 0.948 recall and 0.47 false skips while requiring 672.8 minutes of instruction. Compared with Joint Gap at $\alpha = 0.7$, this reduces the recommended study time by 56.4 minutes with an absolute recall difference of 0.005. The result suggests that targeted assessment can resolve uncertain cases before additional coursework is assigned.

Table 3
Sensitivity of the Joint Gap policy to α; means over 1,000 simulated learners. Assess-Then-Plan is included as reference.

Policy	Instruction (min)	Recall
Joint Gap ($\alpha = 0.00$)	677.6	.836
Joint Gap ($\alpha = 0.25$)	740.7	.942
Joint Gap ($\alpha = 0.50$)	746.4	.960
Joint Gap ($\alpha = 0.70$)	729.2	.953
Joint Gap ($\alpha = 0.75$)	719.3	.947
Joint Gap ($\alpha = 1.00$)	536.2	.796
Assess-Then-Plan	**672.8**	**.948**

Sensitivity to the Joint-Gap Weight We also vary the Joint Gap parameter over $\alpha \in \{0, .25, .50, .70, .75, 1\}$ using the same 1,000-learner simulation. Table 3 shows that the observed trade-off is not specific to $\alpha = .70$. Values of .50 and .70 obtain slightly higher recall than Assess-Then-Plan but require respectively 73.6 and 56.4 additional instructional minutes. At .25 and .75, Joint Gap requires more instruction while achieving lower recall. The two extreme settings reduce instructional time further, but at a substantial cost in recall.

5.3. Limitations

The evaluation is controlled and therefore isolates the behavior of the proposed planning and assessment policies, but several limitations should be considered. First, learner mastery, confidence, and response accuracy are generated from parametric distributions. This allows a controlled comparison across policies, but it does not capture factors such as fatigue, motivation, or changes in performance over time. In addition, the Bloom-based item weights are heuristic rather than psychometrically calibrated, and could be replaced in future work with IRT- or knowledge-tracing-based estimates.

The RQ1 benchmark also assumes that completing a module produces its expected skill coverage. In practice, learning gains are uncertain and may differ substantially across learners. The replanning mechanism described in Section 4.3 can adjust the remaining path when new assessment evidence becomes available, but its behavior still needs to be validated with real learner data. The benchmark further assumes that prerequisite annotations and module durations are correct.

Finally, the experiments start from validated skill-resource associations and therefore do not evaluate the accuracy of semantic skill linking or the quality of the generated diagnostic questions. These components should be assessed separately using expert-annotated resources and assessment items.

6. Conclusion

ESCO-LUCE combines ESCO-based semantic modeling with adaptive curriculum planning. By representing job requirements, module prerequisites, learning resources, and learner states through relations linked to shared ESCO skill identifiers, the framework preserves the external taxonomy while supporting application-specific information. The learner model keeps estimated mastery separate from confidence in the available evidence, allowing the system to distinguish between missing skills and skills that are simply insufficiently verified. In the binary-coverage benchmark, the prerequisite-aware A^* planner recovered minimum-duration paths while reducing node expansions. In the simulation with 1,000 learners, we obtained acceptable recall while reducing recommended instructional time. The results support the use of confidence-aware assessment before assigning additional coursework and show how semantic skill representations can be combined with constrained path planning.

Future work will focus on validating the framework with real learner data, evaluating the quality of skill linking and generated assessment items, and extending the planning model to stochastic learning gains.

Declaration on Generative AI

During the preparation of this work, the authors utilized generative language models to assist with language editing, stylistic refinement, and text condensation. All conceptual modeling, algorithmic designs, experimental evaluations, and conclusions were developed, validated, and approved by the authors.

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